Lección 2

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Plan de trabajo

Teoría de relatividad

– Transformaciones de Lorentz

– Grupo de Lorentz y sus propiedades

– Cinemática Relativista y vectores 4-dimensionales

▶ Nuestro Syllabus:

UNIDAD 1. Teoría de la Relatividad Especial
1.1. Transformaciones de Galileo
1.2. Experimento de Michelson y Morley
1.3. Postulados de Einstein
1.4. Transformaciones de Lorentz, Grupo de Lorentz
1.5. Espacio-tiempo de Minkowski y vectores 4-dimensionales
1.6. Cinemática Relativista
1.7. Dinámica Relativista
1.8. Aplicaciones: la paradoja de los gemelos, efecto Doppler relativista

Aviso

Suspensión de clases 16.03 y 18.03

–**Motivo**: Asistencia al congreso

Postulates of Special Relativity (A. Einstein, 1905)

The principle of relativity

The laws of physics apply (are equivalent) in all **inertial** reference frames

- This principle is valid only in inertial frames. It is NOT valid if we consider non-inertial frame (accelerated, rotating, etc).

The universal speed of light

The speed of light in vacuum is the same in all reference frames.

- Valid only for the speed of light **in vacuum**. Inside matter (dielectric), the speed of light

$$v = c/n < c,$$

and is NOT constant (depends on frame, velocity of dielectric, ...). We'll see a bit later in detail what happens there (and will see in detail Fizeau's experiment).

Corollary

Corollary:

- The Lorentz transformation

$$x' = \cosh\eta\, x - \sinh\eta\, ct, \quad ct' = -\sinh\eta\, x + \cosh\eta\, ct, \quad y' = y, \quad z' = z$$

$$\sinh\eta = \frac{v/c}{\sqrt{1 - v^2/c^2}}$$

relate coordinate and time in different frames (replace Galileo's transformation)

– Time is not absolute, depends on frame

– Distances between points and time intervals between events are not absolute

Control problem

►Lorentz transformations (aka "boost"):

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}},$$
$$ct' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}},$$
$$y' = y, \quad z' = z$$

Inverse Lorentz transformations may be found changing direction of velocity

$$\boldsymbol{v} \to -\boldsymbol{v}$$

:

$$x = \frac{x' + v\,t'}{\sqrt{1 - v^2/c^2}},$$
$$ct = \frac{t' + \frac{v\,x'}{c^2}}{\sqrt{1 - v^2/c^2}},$$
$$y' = y, \quad z' = z$$

Suggest how we can generalize the Lorentz transformation for arbitrary velocity $\boldsymbol{v}$, whose direction is **not** collinear to $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$, $\hat{\boldsymbol{z}}$

Control problem (reply)

Write the Lorentz transformation for arbitrary $\boldsymbol{v}$ not collinear to $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$, $\hat{\boldsymbol{z}}$

We need to split the vector $\boldsymbol{x}$ into parts collinear and orthogonal to $\boldsymbol{v}$,

$$\boldsymbol{x}_{||} = (x \cdot \hat{v})\, \hat{\boldsymbol{v}} = \frac{(x \cdot v)\, \boldsymbol{v}}{v^2},$$
$$\boldsymbol{x}_{\perp} = \boldsymbol{x} - \boldsymbol{x}_{||} = \boldsymbol{x} - \frac{(x \cdot v)\, \boldsymbol{v}}{v^2}$$

▷ Apply Lorentz transformation for collinear and $\perp$ parts:

$$\boldsymbol{x}'_{||} = \gamma\left(\boldsymbol{x}_{||} - \boldsymbol{v}t\right) = \gamma\left(\frac{(x \cdot v)\, \boldsymbol{v}}{v^2} - \boldsymbol{v}t\right)$$
$$\boldsymbol{x}'_{\perp} = \boldsymbol{x}_{\perp} = \boldsymbol{x} - \frac{(x \cdot v)\, \boldsymbol{v}}{v^2}$$

$$\boldsymbol{x}' = \boldsymbol{x}'_{||} + \boldsymbol{x}'_{\perp} = \boldsymbol{x} + \boldsymbol{v}\left[\frac{\boldsymbol{x} \cdot \boldsymbol{v}}{v^2}(\gamma - 1) - \gamma t\right]$$
$$t' = \gamma\left(t - \frac{\boldsymbol{x} \cdot \boldsymbol{v}}{c^2}\right)$$

Lorentz boosts

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \qquad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}}, \quad y' = y, \quad z' = z$$

Generalization for boosts in arbitrary direction:

$$\boldsymbol{x}' = \boldsymbol{x}'_{||} + \boldsymbol{x}'_{\perp} = \boldsymbol{x} + \boldsymbol{v}\left[\frac{\boldsymbol{x}\cdot\boldsymbol{v}}{v^2}(\gamma - 1) - \gamma t\right], \quad t' = \gamma\left(t - \frac{\boldsymbol{x}\cdot\boldsymbol{v}}{c^2}\right)$$

Can rewrite in compact matrix form: $x'^{\mu} = \Lambda^{\mu}{}_{\nu}(\boldsymbol{v})\, x^{\nu}$

$$\Lambda^{\alpha}{}_{\mu}\Lambda_{\alpha}{}^{\nu} = \delta^{\nu}_{\mu}, \qquad \eta_{\alpha\beta}\Lambda^{\alpha}{}_{\mu}\Lambda_{\nu}{}^{\beta} = \eta_{\mu\nu}, \qquad \Lambda_{\mu}{}^{\alpha}(\boldsymbol{v}) = \left[\Lambda^{-1}(\boldsymbol{v})\right]^{\alpha}{}_{\mu} \equiv \Lambda^{\alpha}{}_{\mu}(-\boldsymbol{v})$$

where $\Lambda^{\mu}{}_{\nu}(\boldsymbol{v})$ is the matrix which corresponds to Lorentz boost (akin to rotation matrix for rotations):

$$\Lambda^{\mu}_{\nu} = \begin{pmatrix} \gamma & -v_x\,\gamma & -v_y\,\gamma & -v_z\,\gamma \\ -\frac{v_x\,\gamma}{c^2} & 1+\frac{\bar{\gamma}v_x^2}{v^2} & \frac{\bar{\gamma}v_x v_y}{v^2} & \frac{\bar{\gamma}v_x v_z}{v^2} \\ -\frac{v_y\,\gamma}{c^2} & \frac{\bar{\gamma}v_x v_y}{v^2} & 1+\frac{\bar{\gamma}v_y^2}{v^2} & \frac{\bar{\gamma}v_y v_z}{v^2} \\ -\frac{v_z\,\gamma}{c^2} & \frac{\bar{\gamma}v_x v_z}{v^2} & \frac{\bar{\gamma}v_y v_z}{v^2} & 1+\frac{\bar{\gamma}v_z^2}{v^2} \end{pmatrix},$$

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}, \qquad \bar{\gamma} = \gamma - 1$$

$$\text{for boost in direction } \hat{\boldsymbol{x}}: \qquad \Lambda^{\mu}_{\nu}(v\,\hat{\boldsymbol{x}}) = \begin{pmatrix} \gamma & -v\,\gamma & 0 & 0 \\ -\frac{v\,\gamma}{c^2} & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

- Inverse Lorentz transformations: just change $\boldsymbol{v} \to -\boldsymbol{v}$

Lorentz boosts

$$\Lambda^{\mu}{}_{\nu}(\boldsymbol{v}) = \begin{pmatrix} \gamma & -v_x\,\gamma & -v_y\,\gamma & -v_z\,\gamma \\ -\frac{v_x\,\gamma}{c^2} & 1+\frac{\bar{\gamma}\,v_x^2}{v^2} & \frac{\bar{\gamma}\,v_x\,v_y}{v^2} & \frac{\bar{\gamma}\,v_x\,v_z}{v^2} \\ -\frac{v_y\,\gamma}{c^2} & \frac{\bar{\gamma}\,v_x\,v_y}{v^2} & 1+\frac{\bar{\gamma}\,v_y^2}{v^2} & \frac{\bar{\gamma}\,v_y\,v_z}{v^2} \\ -\frac{v_z\,\gamma}{c^2} & \frac{\bar{\gamma}\,v_x\,v_z}{v^2} & \frac{\bar{\gamma}\,v_y\,v_z}{v^2} & 1+\frac{\bar{\gamma}\,v_z^2}{v^2} \end{pmatrix},$$

$$\gamma = \frac{1}{\sqrt{1-v^2/c^2}}, \qquad \bar{\gamma} = \gamma - 1$$

- If we set $c = 1$ (effectively use variable $x^0 \equiv ct$), can see that for <u>pure boosts</u> the matrix $\Lambda^{\mu}{}_{\nu}(\boldsymbol{v})$ is <u>always</u> symmetric (!).
- Inverse Lorentz transformations: just change $\boldsymbol{v} \to -\boldsymbol{v}$

$$\text{for boost in direction } \hat{\boldsymbol{x}}: \qquad \Lambda^{\mu}_{\nu}(v\,\hat{\boldsymbol{x}}) = \begin{pmatrix} \gamma & -v\,\gamma & 0 & 0 \\ -\frac{v\,\gamma}{c^2} & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\Rightarrow x' = \frac{x - vt}{\sqrt{1-v^2/c^2}}, \qquad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1-v^2/c^2}}, \quad y' = y, \quad z' = z$$

Lemma

Assume that we make two consecutive Lorentz boosts with velocities $v_{1,2}$ in direction $\hat{x}$. The combined transformation is also a boost in direction $\hat{x}$ with rapidity

$$\eta = \eta_1 + \eta_2 = \operatorname{arctanh}\left(\frac{v_1}{c}\right) + \operatorname{arctanh}\left(\frac{v_2}{c}\right)$$

(i.e. rapidities "sum up", as angles in case of rotation around the same axis)

Lemma (proof)

We found earlier that Lorentz transofrmation in direction $\hat{x}$ is given by

$$\begin{aligned} x' &= \cosh\eta\, x - \sinh\eta\, ct \\ ct' &= -\sinh\eta\, x + \cosh\eta\, ct \\ y' &= y, \quad z' = z \end{aligned}$$

Let's disregard coordinates y, z which transform trivially and rewrite this transformation as

$$\begin{pmatrix} x' \\ ct' \end{pmatrix} = \begin{pmatrix} \cosh\eta & -\sinh\eta \\ -\sinh\eta & \cosh\eta \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix}$$

For superposition of 2 boosts have

$$\begin{aligned} \begin{pmatrix} x' \\ ct' \end{pmatrix} &= \begin{pmatrix} \cosh\eta_2 & -\sinh\eta_2 \\ -\sinh\eta_2 & \cosh\eta_2 \end{pmatrix} \begin{pmatrix} \cosh\eta_1 & -\sinh\eta_1 \\ -\sinh\eta_1 & \cosh\eta_1 \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix} \\ &= \begin{pmatrix} \cosh\eta & -\sinh\eta \\ -\sinh\eta & \cosh\eta \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix} \end{aligned}$$

where $\eta = \eta_1 + \eta_2$ and we simply multiplied 2 colored matrices, using

$$\begin{aligned} \cosh(\eta_1 + \eta_2) &= \cosh\eta_1 \cosh\eta_2 + \sinh\eta_1 \sinh\eta_2, \\ \sinh(\eta_1 + \eta_2) &= \sinh\eta_1 \cosh\eta_2 + \cosh\eta_1 \sinh\eta_2, \end{aligned}$$

Control problem 2

Assume that we make two consecutive Lorentz boost in directions $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$. Demonstrate, multiplying Lorentz boost matrices

$$(\Lambda_1)^{\mu}_{\nu} \equiv \Lambda^{\mu}_{\nu}\,(v\,\hat{\boldsymbol{x}}), \quad (\Lambda_2)^{\mu}_{\nu} \equiv \Lambda^{\mu}_{\nu}\,(u\,\hat{\boldsymbol{y}})\,,$$

that the order of multiplication is important, *i.e.* boosts are not commutative ($\Lambda_1 \cdot \Lambda_2 \neq \Lambda_2 \cdot \Lambda_1$). Also, prove that in both cases the result of multiplication ($\Lambda_1 \cdot \Lambda_2$ or $\Lambda_2 \cdot \Lambda_1$) is not pure boost.

Control problem 2 (reply)

▷ Explicit multiplication; use $c = 1$ for simplicity

$$(\Lambda_1)^\mu_\nu = \begin{pmatrix} \gamma(v) & -v\,\gamma(v) & 0 & 0 \\ -v\,\gamma(v) & \gamma(v) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (\Lambda_2)^\mu_\nu = \begin{pmatrix} \gamma(u) & 0 & -u\,\gamma(u) & 0 \\ 0 & 1 & 0 & 0 \\ -u\,\gamma(u) & 0 & \gamma(u) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$(\Lambda_1)^\mu_\alpha\,(\Lambda_2)^\alpha_\nu = \begin{pmatrix} \gamma(u)\gamma(v) & -v\,\gamma(v) & -u\,\gamma(u) & 0 \\ -v\,\gamma(u)\gamma(v) & \gamma(v) & \gamma(u)\gamma(v)uv & 0 \\ -u\,\gamma(u) & 0 & \gamma(u) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$(\Lambda_2)^\mu_\alpha\,(\Lambda_1)^\alpha_\nu = \begin{pmatrix} \gamma(u)\gamma(v) & -v\,\gamma(u)\gamma(v) & -u\,\gamma(u) & 0 \\ -v\,\gamma(v) & \gamma(v) & 0 & 0 \\ -u\,\gamma(u) & \gamma(u)\gamma(v)uv & \gamma(u) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

From pairwise comparison of the same-color terms, can see that $\Lambda_1 \cdot \Lambda_2 \neq \Lambda_2 \cdot \Lambda_1$
From general stucture of Λ^μ_ν expect that for pure boost $\Lambda^\mu_\nu = \Lambda^\nu{}_\mu$. In our case this is NOT so, matrices $\Lambda_1 \cdot \Lambda_2$, $\Lambda_2 \cdot \Lambda_1$ are NOT symmetric

$$\Lambda^i_j = \delta_{ij} + \frac{v_i v_j}{v^2}\,(\gamma - 1)\,,$$

$$\Lambda^i_0 = -\gamma v_i, \quad \Lambda^0_i = \Lambda^i_0,$$

Group axioms

- Mathematics: a set of transformations $g \in G$ is called a group iff:

– The set is closed with respect to multiplication, i.e. $\forall g_1, g_2 \in G$ their product $g_3 = g_1 \cdot g_2 \in G$
– The product is associative. *i.e.*

$$(g_1 \cdot g_2) \cdot g_3 = g_1 \cdot (g_2 \cdot g_3)$$

– There is a unique unit element $e \in G$ such that

$$\forall g \in G, \quad e \cdot g = g \cdot e = g$$

– For any element there is inverse,

$$\forall g \in G, \quad \exists g^{-1} \in G, \quad g \cdot g^{-1} = e$$

In general $g_1 \cdot g_2 \neq g_2 \cdot g_1$, group is NOT commutative. However if for some reason $g_1 \cdot g_2 = g_2 \cdot g_1$, then we say that the group is commutative (abelian).

Lorentz group

As we discussed earlier, if we consider boosts in different directions, then

$$\Lambda(\boldsymbol{v}_1) \cdot \Lambda(\boldsymbol{v}_2)$$

is not a "pure" boost, but rather is a product of boost on rotation matrix

$$\Downarrow$$

"Pure" boosts do not form a group. We should consider both rotations and boosts together. This full set of operations forms a **Lorentz group**, described by (Minkowskian) orthogonal matrices

$$\Lambda^{\mathrm{T}} \cdot \Lambda = 1$$

Prove that all group axioms are satisfied for Lorentz group.

Lorentz group

Proofs:

1) Closeness w.r.t. multiplication:
if $\Lambda_1^{\mathrm{T}} \cdot \Lambda_1 = 1$ and $\Lambda_2^{\mathrm{T}} \cdot \Lambda_2 = 1$, and $\Lambda_3 = \Lambda_1 \cdot \Lambda_2$, then $\Lambda_3^{\mathrm{T}} = (\Lambda_1 \cdot \Lambda_2)^{\mathrm{T}} = \Lambda_2^{\mathrm{T}} \cdot \Lambda_1^{\mathrm{T}}$ and

$$\Lambda_3^{\mathrm{T}} \cdot \Lambda_3 = \Lambda_2^{\mathrm{T}} \cdot \underbrace{\Lambda_1^{\mathrm{T}} \cdot \Lambda_1}_{=1} \cdot \Lambda_2 = \Lambda_2^{\mathrm{T}} \cdot \Lambda_2 = 1$$

2) Associativity is fulfilled automatically for any product of matrices:

$$(A \cdot B) \cdot C = A \cdot (B \cdot C)$$

3) Unit element is

$$\delta^{\mu}_{\nu} \equiv \Lambda^{\mu}{}_{\nu}\left(\boldsymbol{v} = 0\right)$$

4) Inverse element is

$$\left[\Lambda^{-1}\left(\boldsymbol{v}\right)\right]^{\mu}{}_{\nu} = \Lambda^{\mu}{}_{\nu}\left(-\boldsymbol{v}\right)$$

Q.E.D.
NOTE: As discussed, $\Lambda_1 \cdot \Lambda_2 \neq \Lambda_2 \cdot \Lambda_1$, so group is not commutative

Lorentz group

- ▷ Multiplication of boost matrices in general is tedious, difficult for understand group structure
- ▷ Expect: for $\boldsymbol{v} = 0$ $\Lambda^{\mu}_{\nu} = \delta^{\mu}_{\nu}$
- ▷ Let's approximate for $v \ll c$:

$$\Lambda^{\mu}_{\nu}(v) \approx \delta^{\mu}_{\nu} + i \sum_{a} v_a \left(T_a\right)^{\mu}_{\nu} + \mathcal{O}\left(v^2\right)$$

  where $\left(T_a\right)^{\mu}_{\nu}$ are some matrices which do NOT depend on velocity, characterize the Lorentz group.
- ▷ In Mathematics such matrices $\left(T_a\right)^{\mu}_{\nu}$ are called generators, play central role in analysis of the group structure

Generators & their properties

Definition

A derivative of the matrix (operator) $\Lambda(\theta_1, ..., \theta_n)$ with respect to one of the parameters θ_a at point $g = e$ $(\boldsymbol{\theta} = 0)$:

$$T_a = \frac{1}{i} \left. \frac{\partial \Lambda(\theta_1, ..., \theta_n)}{\partial \theta_a} \right|_{\boldsymbol{\theta}=0}$$

– In mathematical books factor $1/i$ is usually omitted; in physical books it is included to have $T_a^\dagger = T_a$ for several physically interesting groups

Why generators matter?

– Close to unit element we may approximate all elements as:

$$\Lambda_1 \approx 1 + i \sum_a \theta_a T_a, \quad \Lambda_2 \approx 1 + i \sum_a \phi_a T_a$$

and simplify group multiplications:

$$\Lambda_1 \cdot \Lambda_2 \approx 1 + i \sum_a (\boldsymbol{\theta} + \boldsymbol{\phi})_a T_a + \mathcal{O}(\theta^2, \phi^2, \theta \cdot \phi)$$

$$\Rightarrow \Lambda_1^{-1} \approx 1 - i \sum_a \theta_a T_a, \quad \Lambda_2^{-1} \approx 1 - i \sum_a \phi_a T_a$$

Generators & their properties

$$\Lambda_1 \approx 1 + i\sum_a \theta_a T_a, \quad \Lambda_2 \approx 1 + i\sum_a \phi_a T_a$$

For **non-commuting groups** $\Lambda_1 \cdot \Lambda_2 \neq \Lambda_2\Lambda_1$

– We can't simply take $\Lambda_1\Lambda_2 - \Lambda_2\Lambda_1$ to study the difference, result might be not boost or rotation at all

– Define commutator as $\Lambda_1\Lambda_2\Lambda_1^{-1}\Lambda_2^{-1}$, get

$$\Lambda_1\Lambda_2\Lambda_1^{-1}\Lambda_2^{-1} = 1 + i^2\,[T_a,\ T_b]\,\theta_a\phi_b + \mathcal{O}(3)$$

*For abelian group $\Lambda_1\Lambda_2\Lambda_1^{-1}\Lambda_2^{-1} = 1$, so $[T_a,\ T_b] = 0$
*For non-abelian group $\Lambda_1\Lambda_2\Lambda_1^{-1}\Lambda_2^{-1} \neq 1$, so $[T_a,\ T_b] \neq 0$
BUT: element $\Lambda_1\Lambda_2\Lambda_1^{-1}\Lambda_2^{-1}$ must be also element of the group (group axiom)

$$\Rightarrow \Lambda_1\Lambda_2\Lambda_1^{-1}\Lambda_2^{-1} \approx 1 + i\lambda_c T_c$$

where λ_c are some (new) parameters (they may depend on θ_a, ϕ_b)
$\Rightarrow$Commutator $[T_a,\ T_b]$ should be some linear combination of T_c,

$$[T_a,\ T_b] = \sum_c i f_{abc} T_c$$

where $\sum_c$ is frequently omitted (Einstein's rule)

Some facts from the theory of Lie groups

▷Definition of the generator:

$$T^a = \frac{1}{i} \left. \frac{\partial \Lambda}{\partial \theta^a} \right|_{\boldsymbol{\theta}=0}$$

where $\boldsymbol{\theta}$ is a set of parameters (veloc-

Lie Algebra

A set of linear operators acting on vector space for which we define a bilinear antisymmetric Lie bracket ("anticommutator")
$[g_1, g_2] = -[g_2, g_1]$ which satisfies the Jacobi identity:

$$[[g_1, g_2], g_3] + [[g_3, g_1], g_2] + [[g_2, g_3], g_1] = 0$$

Lie Algebra (simplified)

A set of commutator relations between the generators

$$[T_a, T_b] = i\, f_{abc}\, T_c$$

where f_{abc} are called structure functions of the group

Lie's theorem

Each Lie algebra $\mathfrak{g}$ corresponds to a simple connected Lie group G whose elements are defined via exponential map. If $\exists$ isomorphism $\varphi : \mathfrak{g} \to \mathfrak{h}$, then there exists a unique homomorphism f of two groups $f : G \to H$

Control problem

Earlier we found that general Lorentz transformation has a form

$$\boldsymbol{x}' = \boldsymbol{x} + \boldsymbol{v}\left[\frac{\boldsymbol{x}\cdot\boldsymbol{v}}{v^2}(\gamma - 1) - \gamma t\right]$$
$$t' = \gamma\left(t - \frac{\boldsymbol{x}\cdot\boldsymbol{v}}{c^2}\right)$$

or equivalently $(x')^\mu = \Lambda^\mu{}_\nu(\boldsymbol{v})x^\nu$ where

$$\Lambda^\mu{}_\nu(\boldsymbol{v}) = \begin{pmatrix} \gamma & -v_x\gamma & -v_y\gamma & -v_z\gamma \\ -\frac{v_x\gamma}{c^2} & 1+\frac{\bar\gamma v_x^2}{v^2} & \frac{\bar\gamma v_x v_y}{v^2} & \frac{\bar\gamma v_x v_z}{v^2} \\ -\frac{v_y\gamma}{c^2} & \frac{\bar\gamma v_x v_y}{v^2} & 1+\frac{\bar\gamma v_y^2}{v^2} & \frac{\bar\gamma v_y v_z}{v^2} \\ -\frac{v_z\gamma}{c^2} & \frac{\bar\gamma v_x v_z}{v^2} & \frac{\bar\gamma v_y v_z}{v^2} & 1+\frac{\bar\gamma v_z^2}{v^2} \end{pmatrix}, \gamma = \frac{1}{\sqrt{1-v^2/c^2}}, \qquad \bar\gamma = \gamma - 1$$

– Write out the infinitesimal transformation ($v \ll c$), keeping only linear $\mathcal{O}(v)$ terms. Identify generators.

– Write out the transformation of coordinates under infinitesimal rotation ($\phi \ll 1$), keeping only linear $\mathcal{O}(\phi)$ terms.

Generators of Lorentz group

►Lorentz boost (use $c = 1$ in this slide):

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu},$$

$$\Lambda^{\mu}_{\nu}(v\,\hat{\boldsymbol{x}}) = \begin{pmatrix} \gamma & -v\,\gamma & 0 & 0 \\ -v\,\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\gamma = \frac{1}{\sqrt{1-v^2}}$$

▷Definition of the generator of the Lie group:

$$T^a = \frac{1}{i}\,\frac{\partial \Lambda}{\partial \alpha^a}\bigg|_{\boldsymbol{\alpha}=0}$$

where $\boldsymbol{\alpha}$ is a set of parameters (velocities and rotation angles in our case)

Generators of Lorentz boosts in directions x, y, z:

$$(K^x)^{\mu}_{\nu} = \begin{pmatrix} 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$(K^y)^{\mu}_{\nu} = \begin{pmatrix} 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$(K^z)^{\mu}_{\nu} = \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix},$$

$$\Rightarrow (K^a)^{\mu}_{\nu} = i\,\left(\delta_{a\mu}\delta_{\nu 0} + \delta_{a\nu}\delta_{\mu 0}\right)$$

Generators of rotations (only spacial indices participate!):

$$(J^a)_{bc} = -i\,\varepsilon_{abc},\ e.g.:\ J_z = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

Introduction: Algebra of rotation group

Some of You may not have seen concept of generator, but I am sure that You already have experience with them.

Recall relations for commutators of angular momentum in Quantum Mechanics:

$$[J^x,\, J^y] = i\, J^z, \quad [J^z,\, J^x] = i\, J^y, \quad [J^y,\, J^z] = i\, J^x,$$

or in compact form

$$[J^a,\, J^b] = i\, \varepsilon_{abc} J^c$$

−Operator of angular momentum ARE generators of rotations, for any function its infinitesimal change is given by

$$f(\boldsymbol{r} + \boldsymbol{\phi} \times \boldsymbol{r}) \approx f(\boldsymbol{r}) + i\boldsymbol{\phi} \cdot \hat{\boldsymbol{J}} f(\boldsymbol{r}) = \left(1 + i\boldsymbol{\phi} \cdot \hat{\boldsymbol{J}}\right) f(\boldsymbol{r})$$

Everything You have seen about angular momentum in QM are actually elements of theory of representations for rotation group.

Algebra of Lorentz group

▷Generators of Lorentz boosts:

$$\Rightarrow (K^z)^\mu_\nu = i\ (\delta_{a\mu}\delta_{\nu 0} + \delta_{a\nu}\delta_{\mu 0})$$

▷Generators of rotations (only spacial indices participate!):

$$(J^a)_{bc} = -i\,\varepsilon_{abc}.$$

Lorentz alegbra ("rules of commutation"):

$$[K^a,\, K^b] = -i\,\varepsilon_{abc} J^c$$

⇒ boosts without rotations do not form closed algebra

$$[K^a,\, J^b] = i\,\varepsilon_{abc} K^c, \quad [J^a,\, J^b] = i\,\varepsilon_{abc} J^c$$

⇒rotations form closed subalgebra ($\mathrm{so}(3)$)

−If we introduce the generators $T_a^{(\pm)} = (J_a \pm iK_a)/2$, the above-given relations (Lie algebra) have a form

$$\left[T_a^{(+)},\, T_b^{(+)}\right] = i\varepsilon_{abc} T_c^{(+)}, \quad \left[T_a^{(-)},\, T_b^{(-)}\right] = i\varepsilon_{abc} T_c^{(-)}, \quad \left[T_a^{(+)},\, T_b^{(-)}\right] = 0$$

i.e. this algebra is isomorphic to $\mathrm{su}(2) \oplus \mathrm{su}(2) \sim \mathrm{so}(3) \oplus \mathrm{so}(3)$

Lorentz Group - finite transformations

►**Consequence of** $[K^a,\, K^b] = -i\,\varepsilon_{abc} J^c$

– Let's multiply boosts in different directions (trivial with *Mathematica*):

$$\Lambda^{\mu}_{\nu}\left(v_x \hat{\boldsymbol{x}}\right)\Lambda^{\nu}_{\alpha}\left(v_y \hat{\boldsymbol{y}}\right) = \begin{pmatrix} \gamma_x\gamma_y & -v_x\,\gamma_x & -v_y\,\gamma_x\gamma_y & 0 \\ -v_x\,\gamma_x & \gamma_x & v_x v_y \gamma_x\gamma_y & 0 \\ -v_y\gamma_y & 0 & \gamma_y & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\Lambda^{\mu}_{\nu}\left(v_y \hat{\boldsymbol{y}}\right)\Lambda^{\nu}_{\alpha}\left(v_x \hat{\boldsymbol{x}}\right) = \begin{pmatrix} \gamma_x\gamma_y & -v_x\,\gamma_x\gamma_y & -v_y\,\gamma_y & 0 \\ -v_x\,\gamma_x & \gamma_x & 0 & 0 \\ -v_y\gamma_x\gamma_y & v_x v_y \gamma_x\gamma_y & \gamma_y & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\gamma_x = \frac{1}{\sqrt{1-v_x^2}}, \quad \gamma_y = \frac{1}{\sqrt{1-v_y^2}}$$

Conclusions:

– boosts in different directions do not commute (similar to rotations)

– Product of two boosts in different directions is NOT a "pure boost" (recall that for "pure" boosts matrix Λ is symmetric)

– In general $\Lambda(\boldsymbol{v}_1)\Lambda(\boldsymbol{v}_2) = \Lambda(\boldsymbol{v}) \cdot \hat{R}$ where $\hat{R}$ is some rotation matrix.

Lorentz Group - explicit transformations

▷ Explicit form of transformation which corresponds to the generators $T_a^{(\pm)} = (J_a \pm iK_a)/2$:

$$\mathcal{T}_x^{\pm}(\alpha\hat{\boldsymbol{x}}) = e^{i\alpha\,(J_x \pm iK_x)}$$

$$[K^a,\, J^b] = i\,\varepsilon_{abc}K^c \Rightarrow [K_x,\, J_x] = 0$$

$$\mathcal{T}_x^{\pm}(\alpha\hat{\boldsymbol{x}}) = e^{i\alpha\,J_x}e^{\mp\alpha\,K_x}$$

-simple product of rotation and boost (although with imaginary "velocity")

$$\mathcal{T}_x^{\pm} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\alpha & \sin\alpha \\ 0 & 0 & -\sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} \cosh(i\alpha) & \pm\sinh(i\alpha) & 0 & 0 \\ \pm\sinh(i\alpha) & \cosh(i\alpha) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} \cos(\alpha) & \pm i\sin(\alpha) & 0 & 0 \\ \pm i\sin(\alpha) & \cos(i\alpha) & 0 & 0 \\ 0 & 0 & \cos\alpha & \sin\alpha \\ 0 & 0 & -\sin\alpha & \cos\alpha \end{pmatrix}$$

-simultaneous rotation in 01 and 23 hyperplanes (assuming we again use $x_0 = ict$). Similar for other generators $\mathcal{T}_y^{\pm}$, $\mathcal{T}_z^{\pm}$

►You may check yourselves that multiplication rules between $\mathcal{T}_x^{+}$, $\mathcal{T}_y^{+}$, $\mathcal{T}_z^{+}$ are the same as for rotation matrices R_x, R_y, R_z (and similarly for $\mathcal{T}_x^{-}$, $\mathcal{T}_y^{-}$, $\mathcal{T}_z^{-}$)

Lorentz group

- **Properties of Lorentz boosts**:
- Boosts + rotations *form a group* $SO(3,1)$

$SO(n)$: *group of rotations in Euclidean* n*-dim. space*

$SO(p,\, q)$ *group of rotations in pseudoeuclidean space. Minkowski space:* $(p,\, q) = (3,\, 1)$

"S" *implies "special"*, $\det(\Lambda) = +1$
"O" *implies "orthogonal"*, $\Lambda^{\mathrm{T}} \cdot \Lambda = 1$

The orthogonal group $O(3)$ includes a subgroup $SO(3)$ and a disjoint coset (set of elements) with $\det\Lambda = -1$ (a product of an inversion of all the coordinates multiplied by elements of $SO(3)$).

Algebra of Lorentz group

$$\left[T_a^{(+)},\ T_b^{(+)}\right] = i\varepsilon_{abc}T_c^{(+)}, \quad \left[T_a^{(-)},\ T_b^{(-)}\right] = i\varepsilon_{abc}T_c^{(-)}, \quad \left[T_a^{(+)},\ T_b^{(-)}\right] = 0$$

Casimir operator

Operator $\hat{\mathcal{C}}$ which commutes with all generators

$$\left[T_a, \hat{\mathcal{C}}\right] = 0$$

Each irreducible representation is fully characterized by eigenvalue of Casimirs.
Number of Casimirs=rank of group

E.g. Casimir for rotation group: $L^2 = L_x^2 + L_y^2 + L_z^2$

►Since Lorentz algebra $\sim \mathrm{so}(3) \oplus \mathrm{so}(3)$, expect that it has two Casimirs:

$$C_\pm = \sum_a \left(T_a^\pm\right)^2 = \frac{\boldsymbol{J}^2 - \boldsymbol{K}^2}{4} \pm i\frac{\boldsymbol{J}\cdot\boldsymbol{K}}{2}$$

⇒Rank of Lorentz group is 2; each irrep of Lorentz algebra is characterized by eigenvalues of two Casimirs.

Lorentz group

For further reading on Lorentz group and its properties I recommend You the book
Arfken, Weber "*Mathematical Methods for Physicists*"
(also we have courses FIS-380, FIS-383 dedicated to the group theory)

Basic transformation properties

Now we are going to formulate how different objects transform under Lorentz transformations (are invariant scalars, vectors or transform in some other way)

– The rules are very simple, we don't need explicit expression for the matrix Λ^{μ}_{ν}, just assume that such matrix exist

Basic transformation properties

Many properties resemble similar properties of 3D vectors:

- Assume that 2 vectors have contravariant components $a^\mu$and b^μ. Then a linear superposition
$$c^\mu = \alpha\, a^\mu + \beta\, b^\mu, \quad \alpha, \beta = \mathrm{const}$$
with arbitrary constants α, β is also a contravariant vector, i.e. transforms as
$$\left(c'\right)^\mu = \Lambda^\mu{}_\nu c^\nu$$
Summation is defined only for objects of the same rank (same number of free Lorentz indices). We can't sum a vector a^μ and scalar, it is "not covariant" (depends on reference frame)
- Assume that a vectors has contravariant components a^μ. Then an object $a_\mu = \eta_{\mu\nu} a^\nu$ transforms as covariant vector (1-form)
$$a^\mu \to \Lambda^\mu{}_\nu(\boldsymbol{v}) a^\nu, \quad a_\mu \to \Lambda_\mu{}^\nu(\boldsymbol{v}) a_\nu, \quad \Lambda^\mu{}_\nu(\boldsymbol{v}) = \Lambda_\mu{}^\nu(-\boldsymbol{v})$$
- Assume that 2 vectors have contravariant components $a^\mu$and b^μ. Then we may construct a direct (tensorial) product
$$T^{\mu\nu} = a^\mu b^\nu$$
which transforms with **two** independent matrices $\Lambda^\mu{}_\nu$
$$\left(T'\right)^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta T^{\alpha\beta}$$

Transformation of tensors (reminder)

- If we have some tensor $T^{\alpha_1\ldots\alpha_n}_{\beta_1\ldots\beta_k}$ (e.g. direct product of co- and contravariant vectors), then we must add $\Lambda^{\mu}_{\ \nu}(\boldsymbol{v})$ for each contravariant Lorentz index, and $\Lambda_{\mu}^{\ \nu}(\boldsymbol{v})$ for each covariant index in transformation law

– Let's consider tensor in which we **contract** over pair of indices (co- and contravariant):

$$T^{\alpha_1\ldots\alpha_n}_{\beta_1\ldots\beta_k} \to \Lambda^{\alpha_1}_{\ \gamma_1}\ldots\Lambda^{\alpha_n}_{\ \gamma_n}\,\Lambda_{\beta_1}^{\ \delta_1}\ldots\Lambda_{\beta_k}^{\ \delta_k}\,T^{\gamma_1\ldots\gamma_n}_{\delta_1\ldots\delta_k}$$

$$T^{\nu\alpha_1\ldots\alpha_n}_{\nu\beta_1\ldots\beta_k} \to \Lambda^{\nu}_{\ \gamma_0}\Lambda^{\alpha_1}_{\ \gamma_1}\ldots\Lambda^{\alpha_n}_{\ \gamma_n}\Lambda_{\nu}^{\ \delta_0}\Lambda_{\beta_1}^{\ \delta_1}\ldots\Lambda_{\beta_k}^{\ \delta_k}\,T^{\gamma_0\gamma_1\ldots\gamma_n}_{\delta_0\delta_1\ldots\delta_k}$$

the two Λ-matrices just cancel in view of orthogonality:

$$\Lambda^{\nu}_{\ \gamma_0}\Lambda_{\nu}^{\ \delta_0} = \delta^{\delta_0}_{\gamma_0}, \qquad T^{\nu\alpha_1\ldots\alpha_n}_{\nu\beta_1\ldots\beta_k} \to \Lambda^{\alpha_1}_{\ \gamma_1}\ldots\Lambda^{\alpha_n}_{\ \gamma_n}\Lambda_{\beta_1}^{\ \delta_1}\ldots\Lambda_{\beta_k}^{\ \delta_k}\,T^{\nu\gamma_1\ldots\gamma_n}_{\nu\delta_1\ldots\delta_k}$$

$\Rightarrow$ After contraction we get tensor of a lower rank $r-2$. For its Lorentz transformation need to add Λ-matrices only for free (uncontracted) Lorentz indices, e.g.: $A^{\mu\nu}b_{\nu}$ will transform as contravariant components of vector a^{μ}, and $a^{\mu}b_{\mu}$ is a scalar.

Representation (basic introduction)

Representation:

- We say that a set of operators $\Lambda(g)$ acting on vectors R forms representation of a group G, if for any vector V from R and for any $g \in G$, the element $V' = \Lambda(g)\, V$ also belongs to R. Representation is irreducible, if it does not include subset which forms representation itself

Rotation group:

- As You remember, in Quantum Mechanics a particle with angular momentum L may have angular momentum projections $L_z \in (-L, L)$. Under rotations all such states mix with each other. This is example that for the same rotation group we have different representations (of dimension $2L+1$, i.e. with $2L+1$ elements)
- If we have vectors $\boldsymbol{a}$ and $\boldsymbol{b}$, we may construct bilinear combinations which do not mix with each other under rotations:

$$a \cdot b, \;\; \boldsymbol{a} \times \boldsymbol{b}, \;\; A_{ij} = a_i b_j + a_j b_i - \frac{2\delta_{ij}}{3}\,(a \cdot b)$$

  These objects are examples of irreps of dimensions 1,3,5 (i.e. objects with 1,3,5 independent components)
- In general in rotation group we may have representation with $2L+1$ elements, where L is integer or semiinteger ("spinor")

Control question

- Earlier result: Lorentz group $\sim SO(3) \times SO(3)$, so all irreps might be characterized by a pair of quantum numbers (p, q) for each subgroup, so the corresponding dimension of irreducible representation is

$$(2p+1)(2q+1)$$

– Identify irreducible representation [values (p, q)] which corresponds to:
  1) Scalar [tensor of rank 0]
  2) Vector a^{μ} [tensor of rank 1]
  3) Tensor $A^{\mu\nu}$ [tensor of rank 2]

Comment on theory of irreps

- From the point of the theory of Lie groups, tensors belong to representation which is a *direct product* of other representations:

$$T^{\alpha_1 \dots \alpha_n}_{\beta_1 \dots \beta_k} \to \Lambda^{\alpha_1}_{\gamma_1} \dots \Lambda^{\alpha_n}_{\gamma_n} \Lambda_{\beta_1}^{\delta_1} \dots \Lambda_{\beta_k}^{\delta_k} T^{\gamma_1 \dots \gamma_n}_{\delta_1 \dots \delta_k}$$

It may be decomposed as a sum of several irreducible representations

Example:

- Assume that we have a rank-2 tensor $T^{\mu\nu}$

– May decompose it into symmetric traceless, antisymmetric and traceful parts:

$$T^{\mu\nu} = S^{\mu\nu} + A^{\mu\nu} + \eta^{\mu\nu} T,$$

$$S^{\mu\nu} = \frac{T^{\mu\nu} + T^{\nu\mu}}{2} - \eta^{\mu\nu} T, \quad A^{\mu\nu} = \frac{T^{\mu\nu} - T^{\nu\mu}}{2}, \quad T = \eta_{\alpha\beta} T^{\alpha\beta}$$

These 3 terms do NOT mix under boosts, i.e. belong to different irreducible representations

Comment on theory of irreps (II)

- Tensors allow to get covariant results, but nevertehless some questions may be understood only using the theory of irreps of Lorentz group:

Example:

–Assume that we have antisymmetric tensor $F_{\mu\nu} = -F_{\nu\mu}$

–We start constructing all possible Lorentz-invariant scalars:

$$\mathrm{tr}\hat{F} = F_{\mu\nu}\eta^{\nu\mu}, \quad \mathrm{tr}\hat{F}^2 = F_{\mu\nu}F^{\nu\mu}, \quad \mathrm{tr}\hat{F}^3 = F_{\mu\nu}F^{\nu\alpha}F_{\alpha\lambda}\eta^{\mu\lambda}, \quad \ldots \det\hat{F}, \ldots$$

–You observe that all Lorentz-invariant terms are functions of only 2 invariants: $I_1 = F_{\mu\nu}F^{\mu\nu}$ and $I_2 = \varepsilon^{\mu\nu\alpha\beta}F_{\mu\nu}F_{\alpha\beta}$:

$$\mathrm{Det}(F) = I_2^2/4, \qquad \mathrm{Tr}\,(F\cdot F) \equiv I_1$$

$$\mathrm{Tr}\,(F\cdot F\cdot F\cdot F) = \frac{I_1^2}{2} - \frac{I_2^2}{4}$$

$$\mathrm{Tr}\,(F\cdot F\cdot F\cdot F\cdot F\cdot F) = \frac{I_1^3}{4} - \frac{3\,I_1\,I_2^2}{16},$$

$$\ldots$$

$$\mathrm{Tr}(\underbrace{F\cdot\ldots\cdot F}_{\text{odd number}}) = 0$$

Why ?

Covariance of equations

The principle of relativity

The laws of physics apply (are equivalent) in all inertial reference frames

$$\Downarrow$$

- An equation is said to be Lorentz covariant if has the same form in any inertial reference frame. To guarantee this, we should use only Lorentz covariant quantities
- Principle of covariance: All physical laws should be formulated using only those physical quantities the measurements of which the observers in different frames of reference could unambiguously correlate.
- =we should NOT mix covariant 4-vectors and 3-vectors

▷**Our goal**: learn how to write equations *covariantly*.

3D vs 4D covariant notations

Caution:

- Levi-Civita ε_{abc} is NOT Lorentz-covariant. Should replace it with 4D $\varepsilon_{\mu\nu\alpha\beta}$ or omit altogether/rewrite in terms of covariant structures.
  - ▷ Such definition is Lorentz-invariant

$$\frac{\partial x^\alpha}{\partial x'^{\mu'}} \frac{\partial x^\beta}{\partial x'^{\nu'}} \frac{\partial x^\gamma}{\partial x'^{\rho'}} \frac{\partial x^\delta}{\partial x'^{\lambda'}} \epsilon_{\alpha\beta\gamma\delta} = \epsilon_{\mu'\nu'\rho'\lambda'} \left| \frac{\partial x^\delta}{\partial x'^{\lambda'}} \right|$$

  and Jacobian $\left|\partial x^\delta / \partial x'^{\lambda'}\right| = 1$ as (recall proof of phase volume invariance)

– Instead of rotations around axis ϕ speak about rotation in plane

– Have *element of a surface (2-form, bivector)*

$$dA^{\mu\nu} = dx^\mu dy^\nu - dy^\mu dx^\nu$$

  formed by vectors dx^μ and dy^ν instead of "$d\boldsymbol{A} = \boldsymbol{n}\,|dA|$ *where* $\boldsymbol{n}$ *is normal to* $d\boldsymbol{x}$, $d\boldsymbol{y}$"

– Bivector ("antisym. tensor of rank 2")

$$A^\mu B^\nu - A^\nu B^\mu$$

  instead of *vector* $\boldsymbol{A} \times \boldsymbol{B}$ (recall that $[\boldsymbol{A} \times \boldsymbol{B}]_3 = A_1 B_2 - A_2 B_1$, ...)

Control question

- Equations of motion (EOM) in noncovariant (3D) notations.

$$\frac{d\boldsymbol{p}}{dt} = \boldsymbol{F} = q\left(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\right) \qquad (1)$$

–

Assume that a pointlike particle moves in external electromagnetic field $(\boldsymbol{E}, \boldsymbol{B})$. Please try to recall (if You remember) how the Lorentz force should be written in covariant notations?

Control question (reply)

- The covariant form is (FIS-221)

$$m\,\frac{du^\mu}{d\tau} \equiv \frac{dp^\mu}{d\tau} = q\,F^{\mu\nu}\,u_\nu \qquad (1)$$

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$A^\mu = (\phi/c,\, \boldsymbol{A})\,, \quad A_\mu = \eta_{\mu\nu} A^\nu = (\phi/c,\, -\boldsymbol{A})\,.$$

$$F^{\mu\nu} = \begin{bmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{bmatrix}$$

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$$

– But we need to revisit how different objects (velocities, time intervals) transform under Lorentz transformations in order to understand why we get such structure

Physical consequence of Lorentz boosts

When You analyze different problems, You should carefully pay attention which transformation should be used: direct or inverse:

$$\Rightarrow x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \qquad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}}, \quad y' = y, \quad z' = z$$

$$\Rightarrow x = \frac{x' + vt'}{\sqrt{1 - v^2/c^2}}, \qquad t = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - v^2/c^2}}, \quad y' = y, \quad z' = z$$

Assume the clock moves with constant velocity v. How the intervals of time measured in rest frame and lab frame are related to each other? How to get this using Lorentz transformation?

Physical consequence of Lorentz boosts

Relativistic time dilation (rest frame vs lab frame):

$$\Delta t_{\mathrm{lab}} = \frac{\Delta t_{\mathrm{r.f.}}}{\sqrt{1 - v^2/c^2}} \equiv \gamma(v)\Delta t_{\mathrm{r.f.}}$$

-time is dilated. Particle physics: for relativistic (*high-energy*) unstable particles their lifetime in lab-frame increases

Example: Muon decay

Assume that a high energy muon was detected to travel a distance L =800 meters in the lab frame before disintegrating. The lifetime of muon in the rest frame is $\tau = 2 \times 10^{-6} s$. According to naive estimate the speed of muon is

$$v = \frac{L}{\tau} = 4 \times 10^8 m/s > c \quad (!!!)$$

Explain the reason of the paradox and estimate the velocity of muon.

Physical consequence of Lorentz boosts

Solution:

In the lab-frame the lifetime of the muon is boosted by factor $\gamma(v)$, so the previous formula should be modified as

$$v = \frac{L}{\tau\gamma(v)} = \frac{L}{\tau}\sqrt{1 - \frac{v^2}{c^2}},$$

so the true velocity is given by

$$v = \frac{L}{\tau\sqrt{1 + L^2/\left(\tau^2 c^2\right)}} \approx 2.4 \times 10^8 m/s \approx 0.8\, c$$

no contradiction with 2nd postulate of Einstein (!)

Control problem - Twin's paradox

(Paul Langevin, 1911)

Assume that the first twin starts a trip in the rocket, whereas the second twin remains in the Earth. The rocket accelerates up to velocity v close to c to achieve a remote star, then turns around and returns back to the Earth. Which of the twins will be older on return ? Analyze how does the picture looks like from the point of view of both twins.

Control problem - Twin's paradox

From the lab on Earth:
(inertial frame)

- ▶The time in the reference frame of the moving twin decelerated by factor $\langle\gamma(v)\rangle$

Assume for simplicity that:

- The distance to the star is d
- Magnitude of velocity is always constant v
- Direction might changes *instantly*.
- From the point of view of the twin on the Earth, the time of flight is

$$T = \frac{2d}{v}$$

▷Since the twin was in a moving frame, his time was Lorentz-dilated for the observer on the Earth, and on arrival he should report that the trip took

$$\tilde{T} = T\sqrt{1-v^2} = T/\gamma.$$

From the rocket's point of view

- ▶The Earth moves with relativistic velocity v
  - distance $d$measured in the Earth's frame is contracted from the point of view of the moving twin,

$$\tilde{d} = d\sqrt{1-v^2} = \frac{d}{\gamma}$$

  - Length of the trip from the point of view of moving observer:

$$\tilde{T} = \frac{2\tilde{d}}{v} = \frac{2d}{v}\sqrt{1-v^2}$$

⇒No contradiction !!!

- ▶More accurate analysis requires to take into account acceleration/deceleration, and the fact that the rocket's reference frame is <u>not</u> inertial ... However final result is the same

Physical consequence of Lorentz boosts

Transformation of space distances depends on orientation:

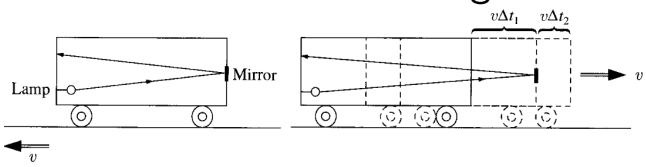
▷We should make *simultaneous* measurement (at the same time) at both ends
▷For transverse coordinates:

$$\Delta y' = \Delta y, \quad \Delta z' = \Delta z$$

▷For logitudinal coordinates:

$$\Delta L_{\rm lab} = \Delta L_{\rm r.f.} \sqrt{1 - v^2/c^2}$$

-Lorentz contraction of longitudinal distances



Assume that we define the length of the train measuring the time required for light to travel forth and back,

$$L = c\,(t_f - t_i)\,/2.$$

Evaluate the length L in different frames and demonstrate that the expressions (A, B) give correct transformation

Solution

▷ **Rest frame:**

–speed of light is c, walls at rest

– time for the ray to reach the mirror and return is

$$t_{\mathrm{r.f.}} = \frac{2L_{\mathrm{r.f.}}}{c} \Rightarrow L_{\mathrm{r.f.}} = \frac{ct_{\mathrm{r.f.}}}{2}$$

▷ **Lab frame:** the velocity of light is c, but:

– wall (mirror) moves away ("escapes") when the ray moves towards mirror, so time to reach the mirror is

$$t_1 = \frac{L_{\mathrm{lab}}}{c - v}$$

– wall (with lamp) moves towards the ray when the latter is reflected, so time to reach the wall is

$$t_2 = \frac{L_{\mathrm{lab}}}{c + v}$$

– Total time in lab frame:

$$t_{\mathrm{lab}} = t_1 + t_2 = \frac{2c\, L_{\mathrm{lab}}}{c^2 - v^2} \qquad (1)$$

– Rest frame time interval $t_{\mathrm{r.f.}}$ is dilated in lab frame,

$$t_{\mathrm{lab}} = \frac{t_{\mathrm{r.f.}}}{\sqrt{1 - v^2/c^2}} = \frac{2L_{\mathrm{r.f.}}}{c\sqrt{1 - v^2/c^2}} \qquad (2)$$

– From comparison of (1,2) conclude $L_{\mathrm{lab}} = L_{\mathrm{r.f.}}\sqrt{1 - v^2/c^2}$ as expected

Einstein's velocity addition

$$x = \frac{x' + Vt'}{\sqrt{1 - V^2/c^2}}, \qquad t = \frac{t' + \frac{Vx'}{c^2}}{\sqrt{1 - V^2/c^2}}, \quad y' = y, \quad z' = z$$

$$x' = \frac{x - Vt}{\sqrt{1 - V^2/c^2}}, \qquad t' = \frac{t - \frac{Vx}{c^2}}{\sqrt{1 - V^2/c^2}}, \quad y' = y, \quad z' = z$$

Assume that the particle moves with velocity $\boldsymbol{v}'$ in some reference frame R', and the reference frame moves in direction $\hat{\boldsymbol{x}}$ of the lab frame with constant velocity V. Find the velocity of particle in the lab frame. Take into account that the velocity $\boldsymbol{v}'$ might have "transverse" components (not collinear to $\hat{\boldsymbol{x}}$).

Einstein's velocity addition

We consider that time is dilated,

$$v_x = \frac{dx}{dt} = \frac{dx' + V\,dt'}{dt' + \frac{V\,dx'}{c^2}} = \frac{v_x' + V}{1 + \frac{v_x' V}{c^2}}$$

$$v_y = \frac{dy}{dt} = \frac{dy'\sqrt{1 - V^2/c^2}}{dt' + \frac{V\,dx'}{c^2}} = \frac{v_y'\sqrt{1 - V^2/c^2}}{1 + \frac{v_x' V}{c^2}}$$

▷For $c \to \infty$

$$v_x \approx v_x' + V, \quad v_y \approx v_y'$$

in agreement with Galileo's velocity addition rule

▷For $v_x' = c$, $v_y' = 0$ we get $v_x = c$, $v_y = 0$ in agreement with 2nd postulate

Einstein's velocity addition

Let's check what happens when light propagates in direction which forms angle $\alpha \neq 0$ with $\hat{\boldsymbol{x}}$:

▷For $v_x' = c\,\cos\alpha$, $v_y' = c\,\sin\alpha$ get

$$v_x = \frac{c\cos\alpha + V}{1 + \frac{V}{c}\cos\alpha}, \quad v_y = \frac{c\sin\alpha\sqrt{1 - V^2/c^2}}{1 + \frac{V}{c}\cos\alpha}$$

$$v_x^2 + v_y^2 = \frac{(c\cos\alpha + V)^2 + c^2\sin^2\alpha\left(1 - \frac{V^2}{c^2}\right)}{\left(1 + \frac{V}{c}\cos\alpha\right)^2} =$$

$$= \frac{\left(c^2 + V^2(1 - \sin^2\alpha) + 2Vc\cos\alpha\right)}{\left(1 + \frac{V}{c}\cos\alpha\right)^2} = c^2$$

–Speed of light in vacuum remains the same (in agreement with 2nd postulate), only changes direction of propagation:

–– Direction of propagation in lab frame <u>changes</u>:

$$\alpha_{\rm lab} = \arctan\left(\frac{v_y}{v_x}\right) = \arccos\left(\frac{v_x}{c}\right) \neq \alpha$$

Control problem 2

- Assume the particle moves with velocity $\boldsymbol{v}'$ in some reference frame R', and the reference frame moves in the lab frame with constant velocity $\boldsymbol{V}$ which does not coincide with $\hat{\boldsymbol{x}}$ What will be the velocity of the particle in the lab frame ?

Control problem 2

$$\boldsymbol{x} = \boldsymbol{x}' + \boldsymbol{V}\left[\frac{\boldsymbol{x}' \cdot \boldsymbol{V}}{V^2}(\gamma - 1) + \gamma t'\right], \quad t = \gamma\left(t' + \frac{\boldsymbol{x}' \cdot \boldsymbol{V}}{c^2}\right), \quad \gamma = \frac{1}{\sqrt{1 - V^2/c^2}}$$

$$\boldsymbol{v} = \frac{d\boldsymbol{x}}{dt} = \frac{\boldsymbol{v}' + \boldsymbol{V}\left[\frac{\boldsymbol{v}' \cdot \boldsymbol{V}}{V^2}(\gamma - 1) + \gamma\right]}{\gamma\left(1 + \frac{\boldsymbol{v}' \cdot \boldsymbol{V}}{c^2}\right)}$$

Einstein's velocity addition

$$v_x = \frac{dx}{dt} = \frac{dx' + V\,dt'}{dt' + \frac{V\,dx'}{c^2}} = \frac{v'_x + V}{1 + \frac{v'_x V}{c^2}}$$

$$v_y = \frac{dy}{dt} = \frac{dy'\sqrt{1 - V^2/c^2}}{dt' + \frac{V\,dx'}{c^2}} = \frac{v'_y\sqrt{1 - V^2/c^2}}{1 + \frac{v'_x V}{c^2}}$$

- The velocity transformation looks complicated (nonlinear expressions).
  -Difficult to work with.

- Most of the complexity (in denominators) due to time interval transformation $dt \to dt'$, dx'

- Assume that now we redefine now all velocities as

$$u^\mu = dx^\mu / d\tau, \qquad (1)$$

where $d\tau$ is the proper time interval (in the rest frame of the partcle)

▷Interval $d\tau$ is:
-Lorentz scalar: time interval in rest frame of a particle
-related to lab frame interval dt as $d\tau = dt\sqrt{1 - v^2/c^2}$

▷All complexity related to denominator is gone. The object (1) will transform as covariant 4-vector (same as coordinates x^μ)

Proper 4-velocity

4-velocity (=Proper velocity)

$$u^\mu \equiv \frac{dx^\mu}{d\tau} = \left(\frac{c}{\sqrt{1 - v^2/c^2}}, \frac{\vec{\boldsymbol{v}}}{\sqrt{1 - v^2/c^2}} \right)$$

- The norm of the 4-velocity vector fulfils

$$u_\mu u^\mu \equiv \eta_{\mu\nu} u^\mu u^\nu = c^2$$

- The 4-velocity transforms as components of: 4-vector (same as for coordinates)

$$u^\mu \to \Lambda^\mu_{\ \nu} u^\nu$$

Special case: boost in $\hat{x}$-direction:

$$\left(u^1\right)' = \gamma \left(u^1 - v u^0\right)$$
$$\left(u^0\right)' = \gamma \left(u^0 - \frac{v u^1}{c^2}\right)$$
$$\left(u^\mu\right)' = \Lambda^\mu_{\ \nu} u^\nu$$

Example:
Assume that a particle moves along $\hat{\boldsymbol{x}}$ direction, and its trajectory is given by

$$x(t) = \sqrt{a^2 + c^2 t^2}, \quad y = z = 0$$

(hyperbolic motion in $x - t$ plane). We'll evaluate the instant velocity $\vec{\boldsymbol{v}}$ and the 4-velocity u^μ for this case

$$v_x = \frac{dx}{dt} = \frac{c^2 t}{\sqrt{a^2 + c^2 t^2}},$$
$$v_y = v_z = 0$$

$$\sqrt{1 - \frac{v^2}{c^2}} \equiv \frac{a}{\sqrt{a^2 + c^2 t^2}}$$

$$u^\mu = \left(\frac{c}{\sqrt{1 - v^2/c^2}}, \frac{\vec{\boldsymbol{v}}}{\sqrt{1 - v^2/c^2}} \right)$$
$$= \frac{c}{a} \left(\sqrt{a^2 + c^2 t^2}, ct \right)$$

Proper 4-momentum

Covariant 4-momentum

$$p^\mu \equiv m\,u^\mu = \left(\frac{mc}{\sqrt{1-v^2/c^2}}, \frac{m\vec{\boldsymbol{v}}}{\sqrt{1-v^2/c^2}}\right),$$

in $v \ll c$ limit

$$\frac{mc}{\sqrt{1-v^2/c^2}} \approx mc\left(1+\frac{v^2}{2\,c^2}\right) \quad (1)$$

$$\approx \frac{1}{c}\left(mc^2 + \underbrace{\frac{mv^2}{2}}_{\text{kinetic energy}}\right)$$

$$\frac{m\vec{\boldsymbol{v}}}{\sqrt{1-v^2/c^2}} \approx \underbrace{m\vec{\boldsymbol{v}}}_{\text{momentum}} \quad (2)$$

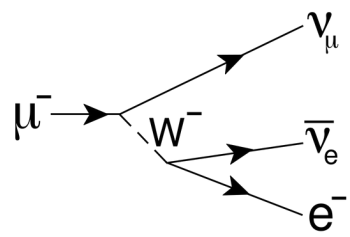
⇒Identify components of p^μ with energy-momentum,

$$p^\mu = \left(\frac{E}{c}, \vec{p}\right), \quad E = \frac{mc^2}{\sqrt{1-v^2/c^2}}$$

- The particle at rest has (huge!) energy

$$E_{\text{rest}} = mc^2$$

- Was strange at the beginning of 20th century, now fully confirmed:
  - Decay of a muon into fragments, $m_\mu \gg \sum m_i$. Rest frame energy converts into kinetic energy of fragments



- Massive particle ($m \neq 0$) cannot be accelerated up to $v \geq c$, energy $E \to \infty$
- The ratio $|p|/E = v/c^2$, so is we assume that some particle might move with $v \approx c$ (*ultrarelativistic limit*), then

$$E = |\vec{p}|\,c$$

Proper 4-momentum

4-momentum (definition)

$$p^\mu \equiv m\, u^\mu = \left(\frac{mc}{\sqrt{1 - v^2/c^2}}, \frac{m\vec{\boldsymbol{v}}}{\sqrt{1 - v^2/c^2}} \right),$$

► By construction the components of the 4-vector p^μ transfrom as coordinates x^μ and 4-velocity u^μ, viz

$$p'^\mu = \Lambda^\mu{}_\nu p^\nu$$

Example: boost in x-direction (replace $ct \to E/c$, $\vec{\boldsymbol{x}} \to \vec{\boldsymbol{p}}$):

$$E' = \frac{E - V\, p_x}{\sqrt{1 - V^2/c^2}}$$

$$p'_x = \frac{p_x - V\, E/c^2}{\sqrt{1 - V^2/c^2}}$$

$$p'_y = p_y, \quad p'_z = p_z$$

4-force

Assume that the force $\vec{F}$ acting on the particle is constant in the laboratory frame. Evaluate the trajectory of the particle. Assume that at $t = 0$ the particle is at rest. Compare your solution with classical result in the limit $v \ll c$

- Choose $\hat{x}$ in direction of $\boldsymbol{F}$
- We work in the lab frame, so

$$F = \frac{dp_x}{dt} \Rightarrow p_x = F\,t + \underbrace{\text{const}}_{=0}$$

$$p_x = \frac{mv}{\sqrt{1 - v^2/c^2}} = F\,t$$

$$v = \frac{F\,t/m}{\sqrt{1 + (F\,t/mc)^2}}, \qquad \lim_{t\to\infty} v(t) = c$$

$$x(t) = \int^t d\tau\, v(\tau) = \frac{mc^2}{F}\left[\sqrt{1 + (F\,t/mc)^2} - 1\right]$$

-hyperbolic motion

▷Case $\lim_{c\to\infty}$, assume $F\,t/mc \ll 1$

$$x(t) \approx \frac{F\,t^2}{2m}$$

in agreement with classical result

Transformation law for the 3-force

Lorentz boosts:

$$x = \frac{x' + Vt'}{\sqrt{1 - V^2/c^2}}, \qquad t = \frac{t' + \frac{Vx'}{c^2}}{\sqrt{1 - V^2/c^2}}, \quad y' = y, \quad z' = z$$

$$x' = \frac{x - Vt}{\sqrt{1 - V^2/c^2}}, \qquad t' = \frac{t - \frac{Vx}{c^2}}{\sqrt{1 - V^2/c^2}}, \quad y' = y, \quad z' = z$$

Transformation for the force is quite sophisticated:

$$F'_y = \frac{dp'_y}{dt'} = \frac{F_y}{1 - v_x\, v_{\mathrm{body}}/c^2}$$

$$F'_z = \frac{dp'_z}{dt'} = \frac{F_z}{1 - v_x\, v_{\mathrm{body}}/c^2}$$

$$F'_x = \frac{dp'_x}{dt'} = \frac{F_x - (v_x/c)\,(dE/dt)}{1 - v_x\, v_{\mathrm{body}}/c^2} = \frac{F_x - (v_x/c)\,(F \cdot v)}{1 - v_x\, v_{\mathrm{body}}/c^2}$$

4-force

Assume that we defined a 4-vector of force

$$g^\mu = dp^\mu / d\tau$$

Evaluate its components in terms of the 3D force $\boldsymbol{f} = d\vec{\boldsymbol{p}}/dt$. Demonstrate that $u \cdot g \equiv u_\mu g^\mu \equiv \eta_{\mu\nu} u^\mu g^\nu = 0$.

$$g^\mu = \left(\frac{\boldsymbol{f} \cdot \boldsymbol{v}}{c\sqrt{1 - v^2/c^2}}, \frac{\boldsymbol{f}}{\sqrt{1 - v^2/c^2}} \right)$$

– Difference from $d\boldsymbol{p}/dt$ by factor $dt/d\tau = 1/\sqrt{1 - v^2/c^2}$
– Recall that $\boldsymbol{f} \cdot \boldsymbol{v}$ is the work done per unit fo time, so we understand why it is 0th component.

$$\eta_{\mu\nu} u^\mu u^\nu = 1,$$

Apply $d/d\tau$ to both parts, recall that $p^\mu = m\, u^\mu$

$$\Rightarrow u \cdot g = 0$$

4-force

►In what follows we may write equations of motion (EOM) in lab frame, using non-covariant derivative d/dt and force

$$\frac{d\boldsymbol{p}}{dt} = \boldsymbol{F} \quad (1)$$

or use covariant Minkowski 4-force $dp^\mu/d\tau$,

$$\frac{dp^\mu}{d\tau} = f^\mu_{\text{external}} \quad (2)$$

But we should always cross-check that both parts of EOM transform *covariantly* (in the same way).

►Example: Lorentz force:

$$\vec{\boldsymbol{F}} = q\left(\vec{\boldsymbol{E}} + \vec{\boldsymbol{v}} \times \vec{\boldsymbol{B}}\right)$$

-is 3-vector (should use (1)), but we may rewrite it as

$$f^\mu = q\, F^{\mu\nu} u_\nu$$

where $F_{\mu\nu}$ is the tensor of electromagetic field and u_ν is the 4-vector

$3D \to 4D$ dictionary

- Vector algebra operations in 4D:

	3D	**4D**
vector	$\vec{\boldsymbol{a}}$	$a^\mu,\ a_\mu = \eta_{\mu\nu} a^\nu$
scalar product	$\boldsymbol{a} \cdot \boldsymbol{b}$	$\eta_{\mu\nu} a^\mu b^\nu \equiv a_\mu b^\mu$
vector product	$(\boldsymbol{a} \times \boldsymbol{b})_i = \frac{\varepsilon_{ijk}}{2} (a_j b_k - a_k b_j)$	$a_\mu b_\nu - a_\nu b_\mu$
gradient	$\vec{\boldsymbol{\nabla}} f$	$\partial_\mu f$
div	$\nabla \cdot \vec{\boldsymbol{a}} \equiv \partial_k a^k$	$\partial_\mu f^\mu \equiv \eta^{\mu\nu} \partial_\mu f_\nu$
rot	$\left(\vec{\nabla} \times \vec{\boldsymbol{a}}\right)_i = \varepsilon_{ijk} \partial_j a_k$ $= \frac{1}{2} \varepsilon_{ijk} (\partial_j a_k - \partial_k a_j)$	$\partial_\mu a_\nu - \partial_\nu a_\mu$
Laplacian	$\Delta \equiv \nabla^2 \equiv \partial_k^2$	$\Box \equiv \partial_\mu \partial^\mu = c^{-2} \partial_t^2 - \Delta$

(see Goldstein, chapter 7 from our bibliography for more details)

Caution with relativistic system

- Relativistic system is not just modification of energy/momentum

►some qualitative features are *different* from $v \ll c$ limit

- We no longer have long-distance interaction

– no $U(\boldsymbol{r}_1 - \boldsymbol{r}_2)$-type terms are allowed in the action

–For closed system should consider interaction *in the same point* x^μ, field+particle.

– MUST include terms both for particles and fields to keep covariance

►Field (e.g. electromagnetic) carries away part of energy-momentum, will affect balance of energy-momentum

Can't introduce concept of center-of-mass

$$\vec{R}_{\mathrm{CM}}(t) = \frac{\sum_{i=1}^{N} m_i \, \vec{x}_i(t)}{\sum_{i=1}^{N} m_i}$$

and "separate" its motion

►Collective momentum $\boldsymbol{P} = \sum \boldsymbol{p}_i$ has nothing to do with $d\vec{R}_{\mathrm{CM}}(t)/dt$.

►We can choose "C-frame" in which $\boldsymbol{P} = 0$, but it is not CM frame!